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Tucker et al.

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(54) **ELECTRONIC VAPING DEVICE INCLUDING
TRANSFER PAD WITH ORIENTED FIBERS**

(58) **Field of Classification Search**

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(Continued)

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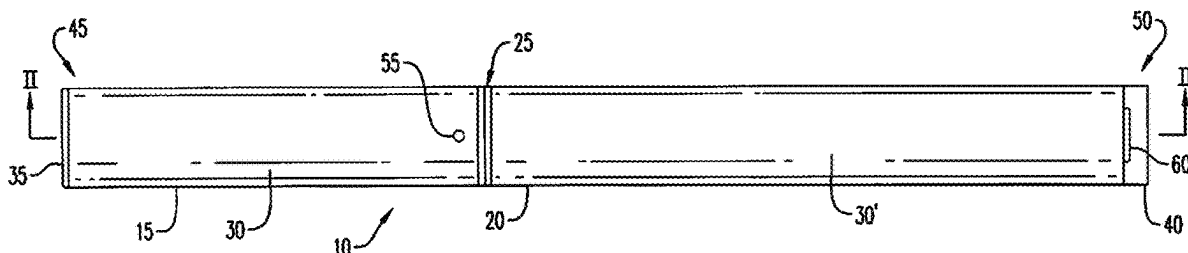
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ABSTRACT

A cartridge for an electronic vaping device includes an outer
housing extending in a longitudinal direction and a reservoir
configured to contain a pre-vapor formulation. The reservoir
is in the outer housing. The cartridge also includes at least
two transfer pads at a reservoir end, and a wick in contact
with the transfer pads. The transfer pads include a plurality
of fibers. Each of the plurality of fibers is substantially
parallel to the longitudinal direction.

33 Claims, 9 Drawing Sheets



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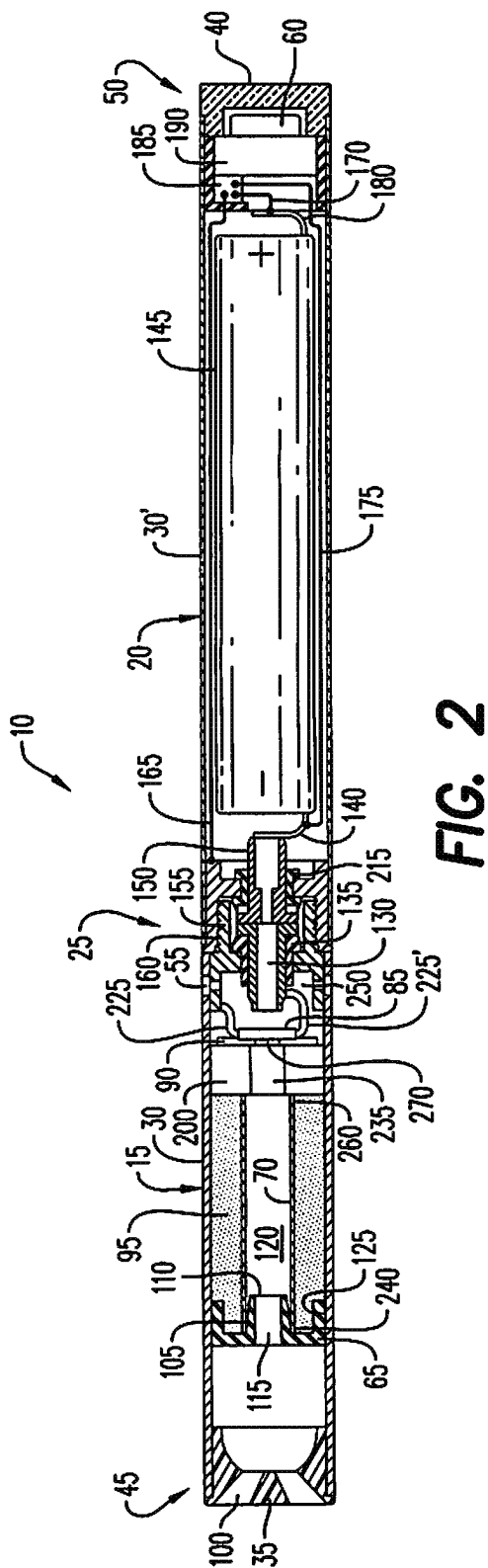
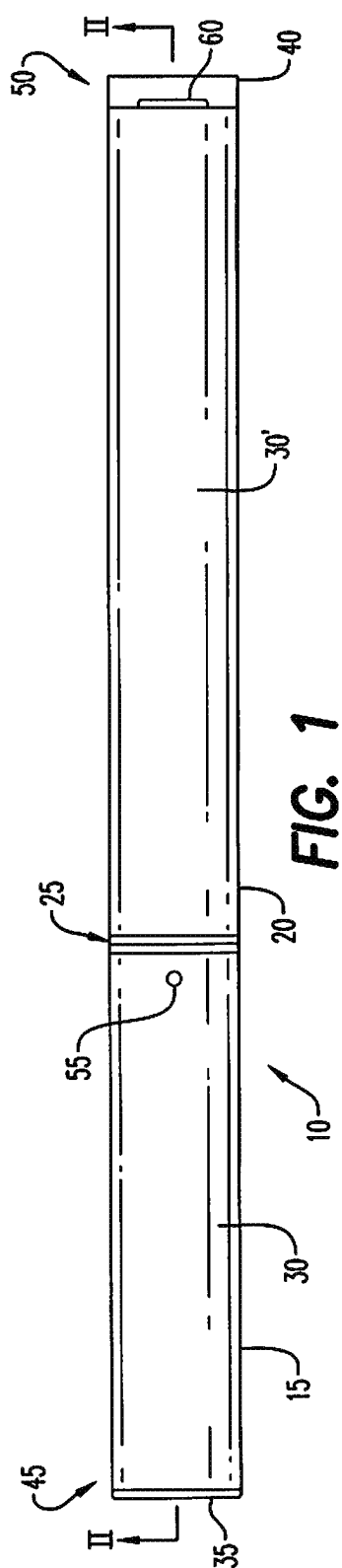
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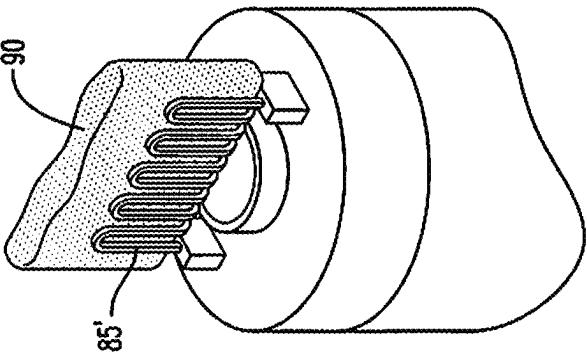
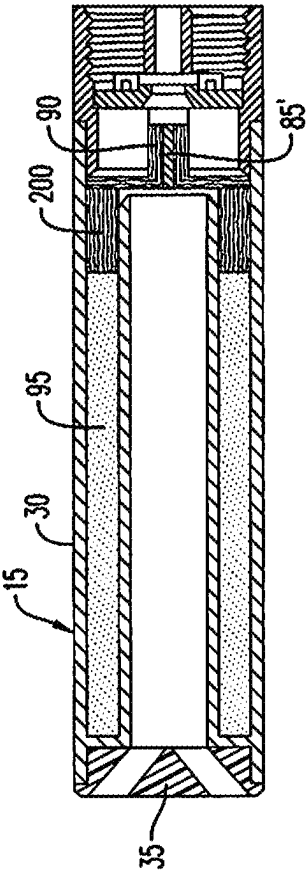
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 Notice of Allowance dated Jun. 17, 2024 issued in Japanese Patent Application No. 2020-517203.
 Decision to Refuse dated Jun. 27, 2024 issued in European patent application No. 19752663.5.
 Notice of Allowance issued Jul. 26, 2024 in U.S. Appl. No. 16/992,810.
 Notice of Allowance dated Jul. 22, 2024 issued in U.S. Appl. No. 15/729,909.
 Office Action dated Aug. 7, 2024 issued in U.S. Appl. No. 18/411,538.

Office Action dated Aug. 7, 2024 issued in U.S. Appl. No. 16/992,810.
 Notice of Allowance dated Aug. 7, 2024 issued in U.S. Appl. No. 16/049,346.
 Notice of Allowance dated Jul. 24, 2024 issued in Korean patent application No. 10-2020-7012756.
 Result of Consultation dated Jun. 5, 2024 issued in European Patent Application No. 19752663.5.
 Notice of Allowance dated Jun. 20, 2024 issued in Chinese Patent Application No. 20188060418.8.
 Office Action mailed Aug. 20, 2024 issued in Brazilian Application No. 1120200052930.
 Notice of Allowance dated Sep. 18, 2024 issued in U.S. Appl. No. 16/049,346.
 Notice of Allowance dated Aug. 28, 2024 issued in U.S. Appl. No. 15/729,909.
 Office Action dated Sep. 26, 2024 issued in U.S. Appl. No. 18/480,586.
 Notice of Allowance dated Oct. 1, 2024 issued in U.S. Appl. No. 17/893,614.
 Office Action dated Sep. 14, 2024 issued in Korean patent application No. 10-2021-7004399.
 Notice of Allowance dated Nov. 21, 2024 issued in U.S. Appl. No. 15/729,909.
 Notice of Allowance dated Nov. 27, 2024 issued in U.S. Appl. No. 16/049,346.
 Notice of Allowance dated Nov. 29, 2024 issued in U.S. Appl. No. 16/992,810.
 Extended European Search Report for European Patent Application No. 24196619.1 issued on Dec. 9, 2024.
 Notice of Allowance dated Dec. 18, 2024 issued in U.S. Appl. No. 18/411,538.
 Notice of Allowance dated Dec. 20, 2024 issued in U.S. Appl. No. 18/411,538.
 Corrected Notice of Allowability dated Jan. 22, 2025 issued in U.S. Appl. No. 16/049,346.
 Office Action dated Jan. 13, 2025 issued in U.S. Appl. No. 18/480,586.
 Notice of Allowance dated Feb. 8, 2025 issued in Korean patent application No. 10-2021-7004862.
 Notice of Allowance dated Feb. 24, 2025 issued in Korean patent application No. 10-2021-7004399.
 Notice of Allowance dated Mar. 11, 2025 issued in U.S. Appl. No. 15/411,538.
 Notice of Allowance dated Mar. 25, 2025 issued in U.S. Appl. No. 16/992,810.
 Notice of Allowance dated Mar. 24, 2025 issued in U.S. Appl. No. 17/893,614.

* cited by examiner





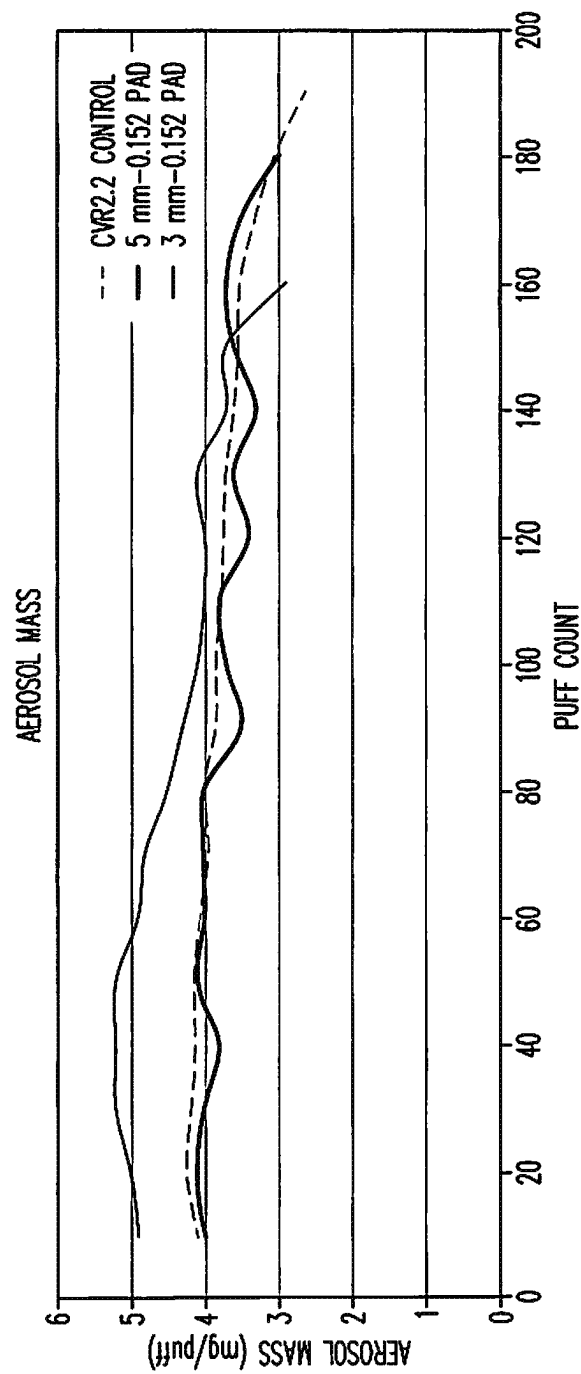


FIG. 4

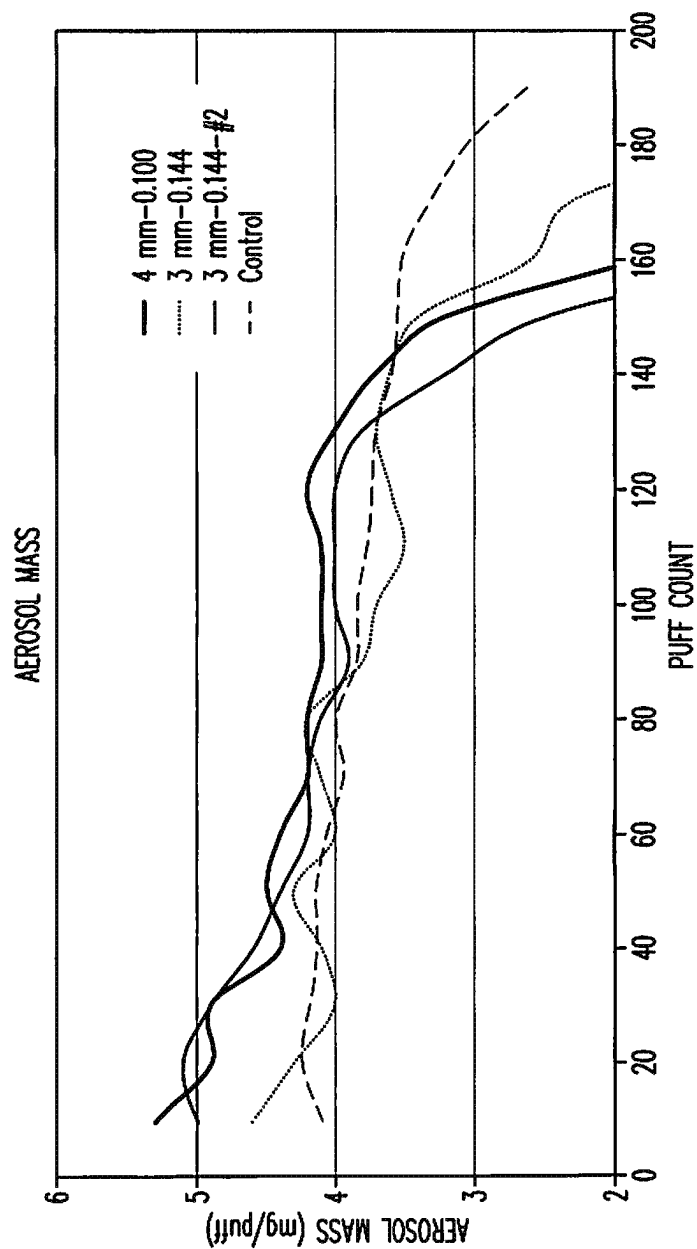


FIG. 5

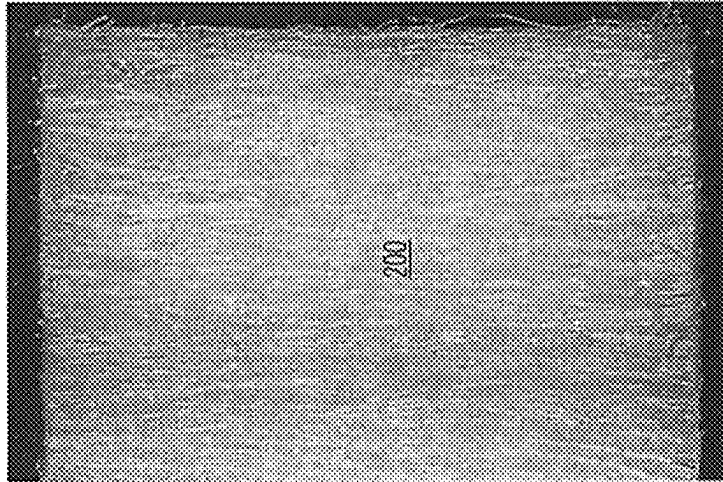


FIG. 7

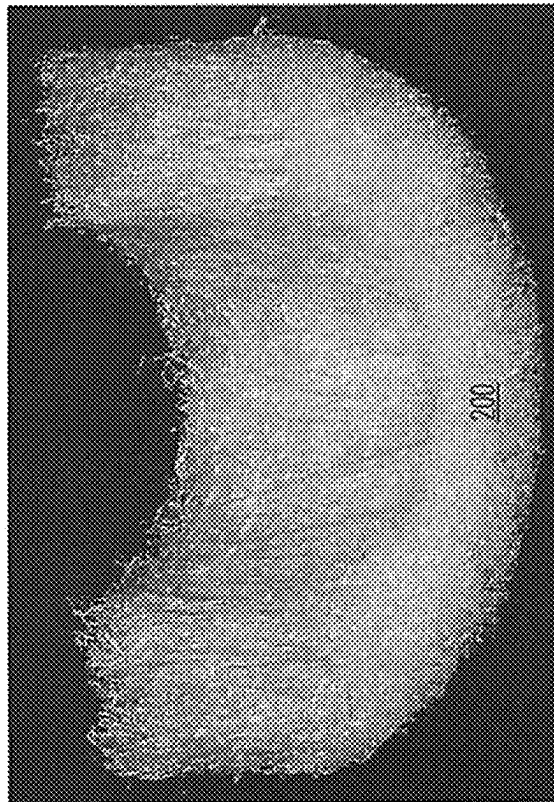


FIG. 6

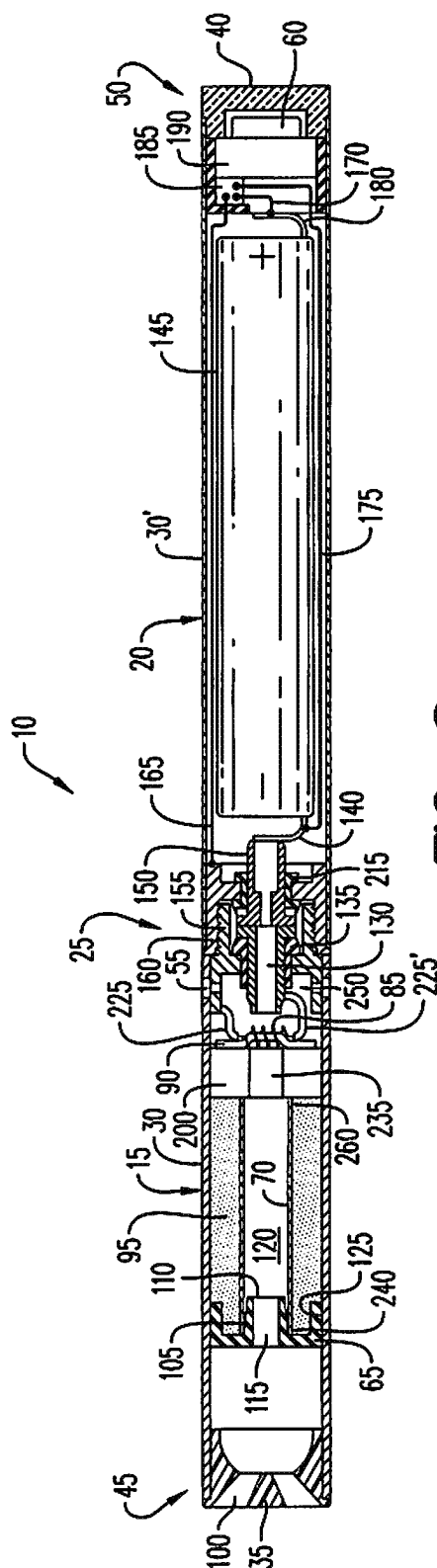


FIG. 8

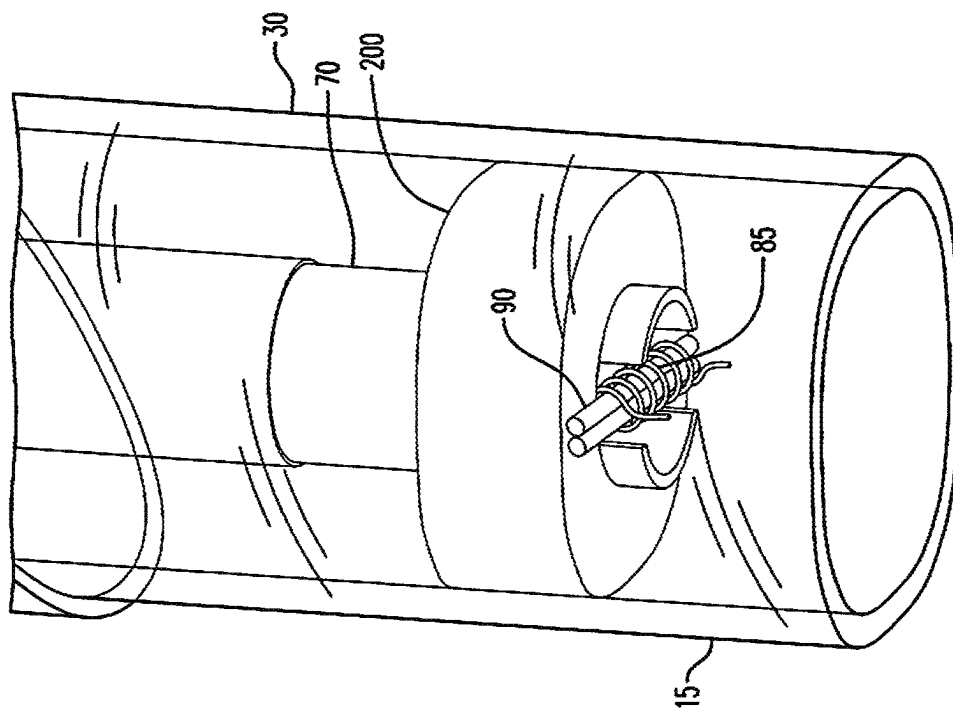


FIG. 9

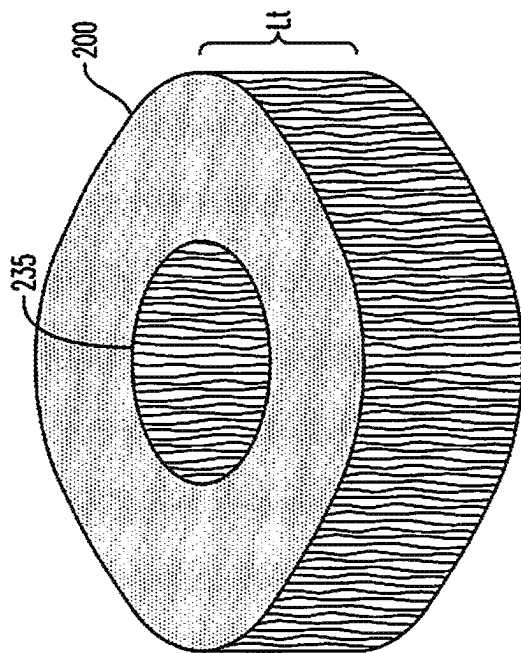


FIG. 10

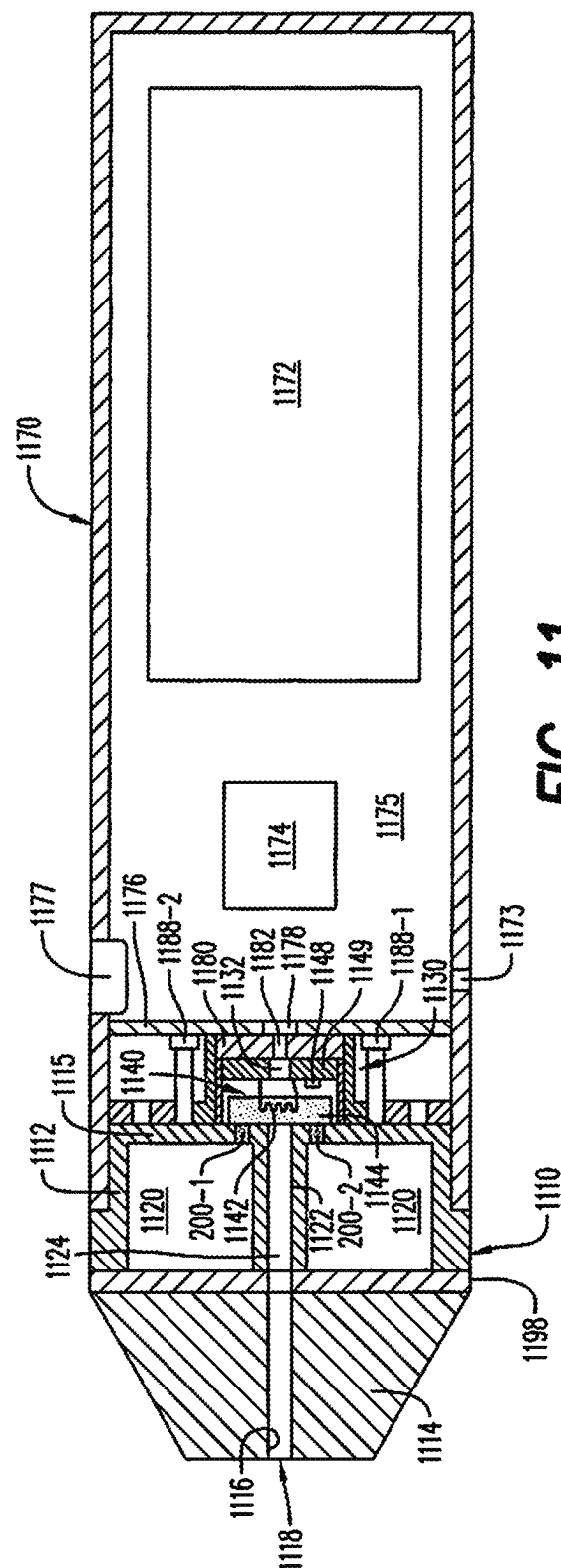


FIG. 11

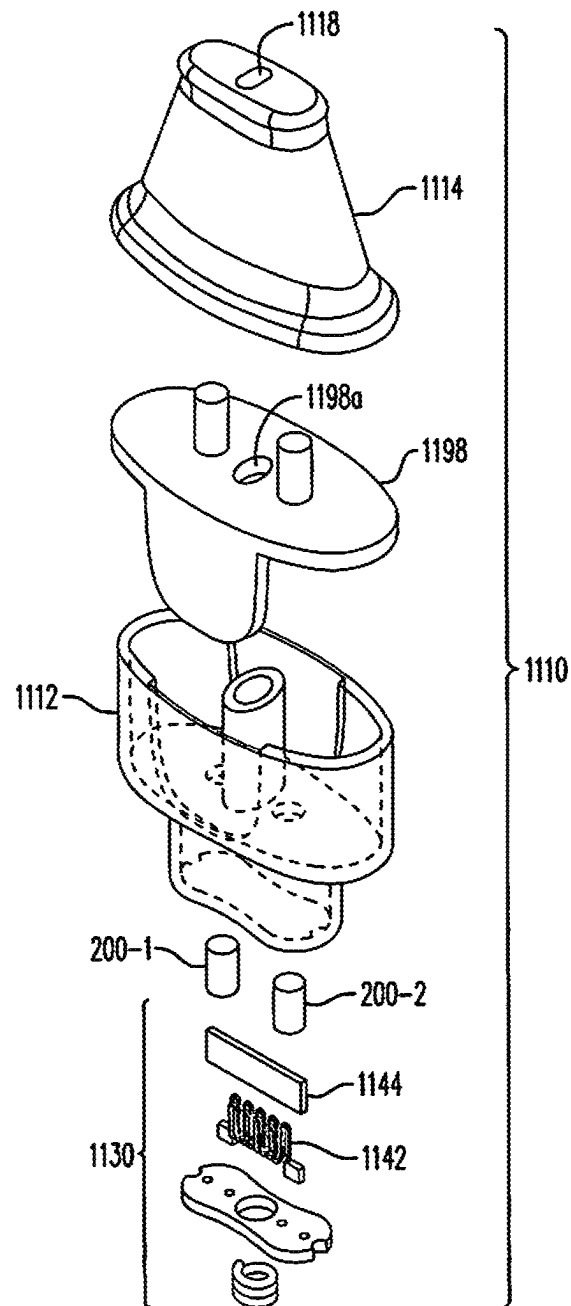


FIG. 12

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**ELECTRONIC VAPING DEVICE INCLUDING
TRANSFER PAD WITH ORIENTED FIBERS****CROSS-REFERENCE TO RELATED
APPLICATION**

This application is a continuation-in-part of U.S. application Ser. No. 15/729,895, filed Oct. 11, 2017, the entire content of which is hereby incorporated by reference.

BACKGROUND**Field**

The present disclosure relates to an electronic vaping or e-vaping device.

Description of Related Art

An e-vaping device includes a heater element which vaporizes a pre-vapor formulation to produce a “vapor.”

The e-vaping device includes a power supply, such as a rechargeable battery, arranged in the device. The battery is electrically connected to the heater, such that the heater heats to a temperature sufficient to convert a pre-vapor formulation to a vapor. The vapor exits the e-vaping device through a mouthpiece including at least one outlet.

SUMMARY

At least one example embodiment relates to a cartridge of an electronic vaping device.

In at least one example embodiment, a cartridge for an electronic vaping device comprises an outer housing extending in a longitudinal direction and a reservoir configured to contain a pre-vapor formulation. The reservoir is in the outer housing. The reservoir has a first reservoir end and a second reservoir end. The cartridge also includes a seal at the first reservoir end, a transfer pad at the second reservoir end, and a wick in contact with the transfer pad. The transfer pad includes a plurality of fibers. Each of the plurality of fibers is substantially parallel to the longitudinal direction.

In at least one example embodiment, the reservoir is under atmospheric pressure.

In at least one example embodiment, the transfer pad has a density ranging from about 0.08 g/cc to about 0.3 g/cc.

In at least one example embodiment, the transfer pad has a length of about 5.0 mm to about 10.0 mm and a density of about 0.08 g/cc to about 0.1 g/cc.

In at least one example embodiment, the transfer pad has a length of about 0.5 mm to about 5.0 mm and a density of about 0.1 g/cc to about 0.3 g/cc.

In at least one example embodiment, the transfer pad includes a plurality of channels. Each of the plurality of channels is between adjacent ones of the plurality of fibers.

In at least one example embodiment, the transfer pad includes an outer side wall. The outer side wall has a coating thereon.

In at least one example embodiment, the transfer pad has a length ranging from about 0.5 mm to about 10.0 mm.

In at least one example embodiment, the cartridge further includes an inner tube within the outer housing. The reservoir is between an outer surface of the inner tube and an inner surface of the outer housing.

In at least one example embodiment, the transfer pad defines a channel extending through the transfer pad. The

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channel is sized and configured to fit around the outer surface of the inner tube at the second end of the reservoir.

In at least one example embodiment, the plurality of fibers includes at least one of polypropylene and polyester.

5 In at least one example embodiment, the cartridge further includes at least one heater in fluid communication with the wick. The at least one heater does not contact the transfer pad.

10 In at least one example embodiment, about 50% to about 100% of the plurality of fibers extend substantially in the longitudinal direction. In at least one example embodiment, about 75% to about 95% of the plurality of fibers extend substantially in the longitudinal direction.

15 At least one example embodiment relates to an electronic vaping device.

20 In at least one example embodiment, an electronic vaping device comprises an outer housing extending in a longitudinal direction and a reservoir configured to contain a pre-vapor formulation. The reservoir is in the outer housing, and the reservoir has a first reservoir end and a second reservoir end. The electronic vaping device also includes a seal at the first reservoir end, a transfer pad at the second reservoir end, a wick in contact with the transfer pad, at least one heater in fluid communication with the wick, and a power supply electrically connectable to the at least one heater. The transfer pad includes a plurality of fibers. The plurality of fibers is substantially parallel to the longitudinal direction.

25 In at least one example embodiment, the reservoir is under atmospheric pressure.

30 In at least one example embodiment, the transfer pad has a density ranging from about 0.08 g/cc to about 0.3 g/cc.

35 In at least one example embodiment, the transfer pad has a length of about 5.0 mm to about 10.0 mm and a density of about 0.08 g/cc to about 0.1 g/cc.

40 In at least one example embodiment, the transfer pad has a length of about 0.5 mm to about 5.0 mm and a density of about 0.1 g/cc to about 0.3 g/cc.

45 In at least one example embodiment, the transfer pad includes a plurality of channels. Each of the plurality of channels is between adjacent ones of the plurality of fibers.

50 In at least one example embodiment, the transfer pad includes an outer side wall. The outer side wall has a coating thereon.

55 In at least one example embodiment, the transfer pad has a length ranging from about 0.5 mm to about 10.0 mm.

60 In at least one example embodiment, the electronic vaping further comprises an inner tube within the outer housing. The reservoir is between an outer surface of the inner tube and an inner surface of the outer housing. The transfer pad is between the outer surface of the inner tube and the inner surface of the outer housing.

65 In at least one example embodiment, the transfer pad defines a channel extending through the transfer pad, and the channel is sized and configured to fit around the outer surface of the inner tube at the second end of the reservoir.

In at least one example embodiment, the plurality of fibers includes at least one of polypropylene and polyester.

In at least one example embodiment, the at least one heater does not contact the transfer pad.

In at least one example embodiment, about 50% to about 100% of the plurality of fibers extend substantially in the longitudinal direction. In at least one example embodiment, about 75% to about 95% of the plurality of fibers extend substantially in the longitudinal direction.

At least one example embodiment relates to a method of forming a cartridge of an electronic vaping device.

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In at least one example embodiment, a method of forming a cartridge of an electronic vaping device comprises positioning an inner tube within an outer housing to establish a reservoir between an outer surface of the inner tube and an inner surface of the outer housing, the inner tube defining an air passage therethrough; inserting a gasket at a first end of the inner tube, the gasket defining a channel in communication with the air passage, the gasket sealing a first reservoir end; and positioning a transfer pad at a second end of the inner tube, the transfer pad including a plurality of fibers, the plurality of fibers being substantially parallel to the longitudinal direction.

In at least one example embodiment, the transfer pad has an outer diameter that is larger than an inner diameter of the outer housing.

In at least one example embodiment, the transfer pad is formed by a melt blowing process.

In at least one example embodiment, the method further comprises positioning a mouth-end insert at a first end of the outer housing.

In at least one example embodiment, the method also includes positioning a wick in contact with the transfer pad; and positioning a heater in contact with the wick, the heater not in physical contact with the transfer pad.

BRIEF DESCRIPTION OF THE DRAWINGS

The various features and advantages of the non-limiting embodiments herein may become more apparent upon review of the detailed description in conjunction with the accompanying drawings. The accompanying drawings are merely provided for illustrative purposes and should not be interpreted to limit the scope of the claims. The accompanying drawings are not to be considered as drawn to scale unless explicitly noted. For purposes of clarity, various dimensions of the drawings may have been exaggerated.

FIG. 1 is a side view of an electronic vaping device according to at least one example embodiment.

FIG. 2 is a cross-sectional view along line II-II of the electronic vaping device of FIG. 1 according to at least one example embodiment.

FIG. 3A is a cross-sectional view of a cartridge according to at least one example embodiment.

FIG. 3B is a perspective view of a heating element and a wick of the cartridge of FIG. 3A according to at least one example embodiment.

FIG. 4 is a graph comparing electronic vaping devices including transfer pads having different densities and/or lengths according to at least one example embodiment.

FIG. 5 is a graph comparing electronic vaping devices including transfer pads having different densities and/or lengths according to at least one example embodiment.

FIG. 6 is a photograph of cross-section of a transfer pad according to at least one example embodiment.

FIG. 7 is an enlarged photograph of a transfer pad according to at least one example embodiment.

FIG. 8 is a side view of an electronic vaping device according to at least one example embodiment.

FIG. 9 is a perspective view of a transfer pad in a cartridge, a housing the cartridge being transparent, according to at least one example embodiment.

FIG. 10 is a perspective view of the transfer pad of FIG. 9 according to at least one example embodiment.

FIG. 11 is a longitudinal cross-sectional view along line of an electronic vaping device according to at least one example embodiment.

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FIG. 12 is an exploded view of the cartridge of FIG. 11 according to at least one example embodiment.

DETAILED DESCRIPTION OF EXAMPLE EMBODIMENTS

Some detailed example embodiments are disclosed herein. However, specific structural and functional details disclosed herein are merely representative for purposes of describing example embodiments. Example embodiments may, however, be embodied in many alternate forms and should not be construed as limited to only the example embodiments set forth herein.

Accordingly, while example embodiments are capable of various modifications and alternative forms, example embodiments thereof are shown by way of example in the drawings and will herein be described in detail. It should be understood, however, that there is no intent to limit example embodiments to the particular forms disclosed, but to the contrary, example embodiments are to cover all modifications, equivalents, and alternatives falling within the scope of example embodiments. Like numbers refer to like elements throughout the description of the figures.

It should be understood that when an element or layer is referred to as being “on,” “connected to,” “coupled to,” or “covering” another element or layer, it may be directly on, connected to, coupled to, or covering the other element or layer or intervening elements or layers may be present. In contrast, when an element is referred to as being “directly on,” “directly connected to,” or “directly coupled to” another element or layer, there are no intervening elements or layers present. Like numbers refer to like elements throughout the specification. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

It should be understood that, although the terms first, second, third, etc. may be used herein to describe various elements, components, regions, layers and/or sections, these elements, components, regions, layers, and/or sections should not be limited by these terms. These terms are only used to distinguish one element, component, region, layer, or section from another region, layer, or section. Thus, a first element, component, region, layer, or section discussed below could be termed a second element, component, region, layer, or section without departing from the teachings of example embodiments.

Spatially relative terms (e.g., “beneath,” “below,” “lower,” “above,” “upper,” and the like) may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. It should be understood that the spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, elements described as “below” or “beneath” other elements or features would then be oriented “above” the other elements or features. Thus, the term “below” may encompass both an orientation of above and below. The device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein interpreted accordingly.

The terminology used herein is for the purpose of describing various example embodiments only and is not intended to be limiting of example embodiments. As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms

“includes,” “including,” “comprises,” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

Example embodiments are described herein with reference to cross-sectional illustrations that are schematic illustrations of idealized embodiments (and intermediate structures) of example embodiments. As such, variations from the shapes of the illustrations as a result, for example, of manufacturing techniques and/or tolerances, are to be expected. Thus, example embodiments should not be construed as limited to the shapes of regions illustrated herein but are to include deviations in shapes that result, for example, from manufacturing.

Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which example embodiments belong. It will be further understood that terms, including those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

FIG. 1 is a side view of an e-vaping device according to at least one example embodiment.

In at least one example embodiment, as shown in FIG. 1, an electronic vaping device (e-vaping device) **10** may include a replaceable cartridge (or first section) **15** and a reusable battery section (or second section) **20**, which may be coupled together at a threaded connector **25**. It should be appreciated that the connector **25** may be any type of connector, such as a snug-fit, detent, clamp, bayonet, and/or clasp. An air inlet **55** extends through a portion of the connector **25**.

In at least one example embodiment, the connector **25** may be the connector described in U.S. application Ser. No. 15/154,439, filed May 13, 2016, the entire contents of which is incorporated herein by reference thereto. As described in U.S. application Ser. No. 15/154,439, the connector **25** may be formed by a deep drawn process.

In at least one example embodiment, the first section **15** may include a first housing **30** and the second section **20** may include a second housing **30'**. The e-vaping device **10** includes a mouth-end insert **35** at a first end **45**.

In at least one example embodiment, the first housing **30** and the second housing **30'** may have a generally cylindrical cross-section. In other example embodiments, the housings **30** and **30'** may have a generally triangular cross-section along one or more of the first section **15** and the second section **20**. Furthermore, the housings **30** and **30'** may have the same or different cross-section shape, or the same or different size. As discussed herein, the housings **30**, **30'** may also be referred to as outer or main housings.

In at least one example embodiment, the e-vaping device **10** may include an end cap **40** at a second end **50** of the e-vaping device **10**. The e-vaping device **10** also includes a light **60** between the end cap **40** and the first end **45** of the e-vaping device **10**.

FIG. 2 is a cross-sectional view along line II-II of the e-vaping device of FIG. 1.

In at least one example embodiment, as shown in FIG. 2, the first section **15** may include a reservoir **95** configured to store a pre-vapor formulation and a vaporizer **80** that may vaporize the pre-vapor formulation. The vaporizer **80** includes a heating element **85** and a wick **90**. The wick **90**

may draw the pre-vapor formulation from the reservoir **95**. The heating element **85** may be a planar heating element including a filament portion, and the wick **90** may be a disk of wicking material as described in U.S. Patent Publication No. 2016/0309786 to Holtz et al. filed on Apr. 22, 2016, the entire content of which is incorporated herein by reference.

The e-vaping device **10** may include the features set forth in U.S. Patent Application Publication No. 2013/0192623 to Tucker et al. filed Jan. 31, 2013, the entire contents of each of which are incorporated herein by reference thereto. In other example embodiments, the e-vaping device may include the features set forth in U.S. Patent Application Publication No. 2016/0309785 filed Apr. 22, 2016, and/or U.S. Pat. No. 9,289,014 issued Mar. 22, 2016, the entire contents of each of which is incorporated herein by reference thereto.

In at least one example embodiment, the pre-vapor formulation is a material or combination of materials that may be transformed into a vapor. For example, the pre-vapor formulation may be a liquid, solid and/or gel formulation including, but not limited to, water, beads, solvents, active ingredients, ethanol, plant extracts, plant material, natural or artificial flavors, and/or vapor formers such as glycerin and propylene glycol. The plant material may include tobacco and/or non-tobacco plant material.

In at least one example embodiment, the first section **15** may include the housing **30** extending in a longitudinal direction and an inner tube (or chimney) **70** coaxially positioned within the housing **30**. The inner tube **70** has a first end **240** and a second end **260**.

A transfer pad **200** abuts the second end **260** of the inner tube **70**. An outer perimeter of the transfer pad **200** may provide a seal with an interior surface of the housing **30**. The transfer pad **200** reduces and/or prevents leakage of liquid from the reservoir **95** that is established between the inner tube **70** and the housing **30**.

In at least one example embodiment, the transfer pad **200** includes a central, longitudinal air passage **235** defined therein. The air passage **235** is in fluid communication with an inner passage (also referred to as a central channel or central inner passage) **120** defined by the inner tube **70**.

In at least one example embodiment, the transfer pad **200** includes a plurality of fibers. Each of the plurality of fibers is substantially parallel to the longitudinal direction. The transfer pad **200** may be formed of at least one of polypropylene and polyester. In at least one example embodiment, the transfer pad **200** may be formed of polyolefin. Other polymers may be used to form the transfer pad **200**.

The transfer pad **200** may be formed by a melt blowing process, where micro- and/or nano-fibers are formed from at least one polymer that is melted and extruded through small nozzles surrounded by high speed blowing gas and/or air.

The polymers used in the melt blowing process do not include any processing aids, such as antistatics, lubricants, bonding agents, and/or surfactants. Thus, the polymers are substantially pure and the transfer pad **200** is inert to the pre-vapor formulation. In other example embodiments, the polymers may be mixed with processing aids, such as antistatics, lubricants, bonding agents, and/or surfactants. The transfer pad **200** may be obtained from Essentra PLC.

In at least one example embodiment, the transfer pad **200** includes an outer side wall. The outer side wall may have a coating thereon that aids in reducing leakage and/or forming a seal between the transfer pad **200** and an inner surface of the housing **30**. The coating may include a hydrophobic or hydrophilic surface texture and/or be a parylene coating.

While not wishing to be bound by theory, melt blowing polymers to form the transfer pad **200** may provide a melted outer surface that also aids in reducing leakage.

In at least one example embodiment, the transfer pad **200** includes a plurality of channels. Each of the plurality of channels is between adjacent ones of the plurality of fibers. The channels may have a diameter ranging from about 0.01 mm to about 0.3 mm (e.g., about 0.02 mm to about 0.2 mm, about 0.03 mm to about 0.1 mm, about 0.04 mm to about 0.09 mm, or about 0.05 mm to about 0.08 mm).

In at least one example embodiment, about 50% to about 100% (e.g., about 55% to about 95%, about 60% to about 90%, about 65% to about 85%, or about 70% to about 75%) of the plurality of fibers extend substantially in the longitudinal direction. In at least one example embodiment, about 75% to about 95% (e.g., about 80% to about 90% or about 82% to about 88%) of the plurality of fibers extend substantially in the longitudinal direction.

The transfer pad may be generally cylindrical or disc shaped, but the transfer pad is not limited to cylindrical or disc shaped forms and a shape of the transfer pad may depend on a shape of the reservoir and housing. An outer diameter of the transfer pad **200** may range from about 3.0 mm to about 20.0 mm (e.g., about 5.0 mm to about 18.0 mm, about 7.0 mm to about 15.0 mm, about 9.0 mm to about 13.0 mm, or about 10.0 mm to about 12.0 mm). The air passage **235** within the transfer pad **200** may have an inner diameter that is substantially the same as an inner diameter of the inner tube **70**. In at least one example embodiment, the inner diameter of the air passage **235** ranges from about 1.0 mm to about 10.0 mm (about 2.0 mm to about 8.0 mm or about 4.0 mm to about 8.0 mm).

In at least one example embodiment, the transfer pad **200** is oriented, such that the channels mostly transverse to the longitudinal direction of the housing **30**. In other example embodiments, the transfer pad **200** is oriented, such that the channels do not run transverse to the longitudinal direction of the housing **30**.

While not wishing to be bound by theory, it is believed that the pre-vapor formulation travels through the channels, and a diameter of the channels is such that a liquid surface tension and pressurization within the reservoir moves and holds the pre-vapor formulation within the channel without leaking.

In at least one example embodiment, the transfer pad **200** may be in a vaping device that does not include a reservoir in a closed system and/or at atmospheric pressure.

In at least one example embodiment, the reservoir is sealed at a first end thereof and the transfer pad **200** is at a second end thereof.

In at least one example embodiment, the transfer pad **200** has a density ranging from about 0.08 g/cc to about 0.3 g/cc (e.g., about 0.01 g/cc to about 0.25 g/cc or about 0.1 g/cc to about 0.2 g/cc). The transfer pad **200** has a length ranging from about 0.5 millimeter (mm) to about 10.0 mm (e.g., about 1.0 mm to about 9.0 mm, about 2.0 mm to about 8.0 mm, about 3.0 mm to about 7.0 mm, or about 4.0 mm to about 6.0 mm). In at least one example embodiment, as the density of the transfer pad **200** increases, the length of the transfer pad decreases. Thus, transfer pads **200** having lower densities within the above-referenced range may be longer than transfer pads **200** having higher densities.

In at least one example embodiment, the transfer pad **200** has a length of about 5.0 mm to about 10.0 mm and a density of about 0.08 g/cc to about 0.1 g/cc.

In at least one example embodiment, the transfer pad **200** has a length of about 0.5 mm to about 5.0 mm and a density of about 0.1 g/cc to about 0.3 g/cc.

In at least one example embodiment, the density and/or length of the transfer pad **200** is chosen based on the viscosity of a liquid flowing therethrough. Moreover, the density of the transfer pad **200** is chosen based on desired vapor mass, desired flow rate of the pre-vapor formulation flow rate, and the like.

In at least one example embodiment, the first connector piece **155** may include a male threaded section for effecting the connection between the first section **15** and the second section **20**.

In at least one example embodiment, at least two air inlets **55** may be included in the housing **30**. Alternatively, a single air inlet **55** may be included in the housing **30**. Such arrangement allows for placement of the air inlet **55** close to the connector **25** without occlusion by the presence of the first connector piece **155**. This arrangement may also reinforce the area of air inlets **55** to facilitate precise drilling of the air inlets **55**.

In at least one example embodiment, the air inlets **55** may be provided in the connector **25** instead of in the housing **30**. In other example embodiments, the connector **25** may not include threaded portions.

In at least one example embodiment, the at least one air inlet **55** may be formed in the housing **30**, adjacent the connector **25** to reduce and/or minimize the chance of an adult vaper's fingers occluding one of the ports and to control the resistance-to-draw (RTD) during vaping. In at least one example embodiment, the air inlet **55** may be machined into the housing **30** with precision tooling such that their diameters are closely controlled and replicated from one e-vaping device **10** to the next during manufacture.

In at least one example embodiment, the air inlets **55** may be sized and configured such that the e-vaping device **10** has a resistance-to-draw (RTD) in the range of from about 60 mm H₂O to about 150 mm H₂O.

In at least one example embodiment, a nose portion **110** of a gasket **65** may be fitted into a first end portion **105** of the inner tube **70**. An outer perimeter of the gasket **65** may provide a substantially tight seal with an inner surface **125** of the housing **30**. The gasket **65** may include a central channel **115** disposed between the inner passage **120** of the inner tube **70** and the interior of the mouth-end insert **35**, which may transport the vapor from the inner passage **120** to the mouth-end insert **35**. The mouth-end insert **35** includes at least two outlets **100**, which may be located off-axis from the longitudinal axis of the e-vaping device **10**. The outlets **100** may be angled outwardly in relation to the longitudinal axis of the e-vaping device **10**. The outlets **100** may be substantially uniformly distributed about the perimeter of the mouth-end insert **35** so as to substantially uniformly distribute vapor.

In at least one example embodiment, the space defined between the gasket **65**, the transfer pad **200**, the housing **30**, and the inner tube **70** may establish the confines of the reservoir **95**. The reservoir **95** may contain the pre-vapor formulation, and optionally a storage medium (not shown) configured to store the pre-vapor formulation therein. The storage medium may include a winding of cotton gauze or other fibrous material about the inner tube **70**. The reservoir is under atmospheric pressure.

In at least one example embodiment, the reservoir **95** may at least partially surround the inner passage **120**. Thus, the reservoir **95** may at least partially surround the inner passage **120**. The heating element **85** may extend transversely across

the inner passage 120 between opposing portions of the reservoir 95. In some example embodiments, the heater 85 may extend parallel to a longitudinal axis of the inner passage 120.

In at least one example embodiment, the reservoir 95 may be sized and configured to hold enough pre-vapor formulation such that the e-vaping device 10 may be configured for vaping for at least about 200 seconds. Moreover, the e-vaping device 10 may be configured to allow each puff to last a maximum of about 5 seconds.

In at least one example embodiment, the storage medium may be a fibrous material including at least one of cotton, polyethylene, polyester, rayon and combinations thereof. The fibers may have a diameter ranging in size from about 6 microns to about 15 microns (e.g., about 8 microns to about 12 microns or about 9 microns to about 11 microns). The storage medium may be a sintered, porous or foamed material. Also, the fibers may be sized to be irrespirable and may have a cross-section which has a Y-shape, cross shape, clover shape or any other suitable shape. In at least one example embodiment, the reservoir 95 may include a filled tank lacking any storage medium and containing only pre-vapor formulation.

During vaping, pre-vapor formulation may be transferred from the reservoir 95 and/or storage medium to the proximity of the heating element 85 via capillary action of the wick 90, which pulls the pre-vapor formulation from the transfer pad 200. The wick 90 may be a generally tubular, and may define an air channel 270 therethrough. The heating element 85 abuts one end of the wick 90. In other example embodiment, the heating element may at least partially surround a portion of the wick 90. When the heating element 85 is activated, the pre-vapor formulation in the wick 90 may be vaporized by the heating element 85 to form a vapor.

In at least one example embodiment, the wick 90 is in direct physical contact with the transfer pad 200, but the heating element 85 does not directly contact the transfer pad 200. In other example embodiments, the heating element 85 may contact the wick 90 and the transfer pad 200 (not shown).

In at least one example embodiment, the wick 90 includes one or more layers of a sheet of wicking material. The sheet of wicking material may be formed of borosilicate or glass fiber. The sheet of wicking material may be folded and/or the wick 90 includes two or more layers of the sheet of wicking material. The sheet of wicking material may have a thickness ranging from about 0.2 mm to about 2.0 mm (e.g., about 0.3 mm to about 1.75 mm, about 0.5 mm to about 1.5 mm, or about 0.75 mm to about 1.0 mm).

In other example embodiments, the wick 90 may include filaments (or threads) having a capacity to draw the pre-vapor formulation. For example, the wick 90 may be a bundle of glass (or ceramic) filaments, a bundle including a group of windings of glass filaments, etc., all of which arrangements may be capable of drawing pre-vapor formulation via capillary action by interstitial spacings between the filaments. The filaments may be generally aligned in a direction perpendicular (transverse) to the longitudinal direction of the e-vaping device 10. In at least one example embodiment, the wick 90 may include one to eight filament strands, each strand comprising a plurality of glass filaments twisted together. The end portions of the wick 90 may be flexible and foldable into the confines of the reservoir 95. The filaments may have a cross-section that is generally cross-shaped, clover-shaped, Y-shaped, or in any other suitable shape.

In at least one example embodiment, the wick 90 may include any suitable material or combination of materials. Examples of suitable materials may be, but not limited to, glass, ceramic- or graphite-based materials. The wick 90 may have any suitable capillarity drawing action to accommodate pre-vapor formulations having different physical properties such as density, viscosity, surface tension and vapor pressure. The wick 90 may be non-conductive.

In at least one example embodiment, the heating element 85 may include a planar sheet of metal that abuts the wick 90. The planar sheet may extend fully or partially along the length of the wick 90. In some example embodiments, the heating element 85 may not be in contact with the wick 90.

In other example embodiment, not shown, the heating element 85 can be in the form of a wire coil, a ceramic body, a single wire, a cage of resistive wire, or any other suitable form. The heating element 85 may be any heater that is configured to vaporize a pre-vapor formulation.

In at least one example embodiment, the heating element 85 may be formed of any suitable electrically resistive materials. Examples of suitable electrically resistive materials may include, but not limited to, copper, titanium, zirconium, tantalum and metals from the platinum group. Examples of suitable metal alloys include, but not limited to, stainless steel, nickel, cobalt, chromium, aluminum-titanium-zirconium, hafnium, niobium, molybdenum, tantalum, tungsten, tin, gallium, manganese and iron-containing alloys, and super-alloys based on nickel, iron, cobalt, stainless steel. For example, the heating element 85 may be formed of nickel aluminide, a material with a layer of alumina on the surface, iron aluminide and other composite materials, the electrically resistive material may optionally be embedded in, encapsulated or coated with an insulating material or vice-versa, depending on the kinetics of energy transfer and the external physicochemical properties required. The heating element 85 may include at least one material selected from the group consisting of stainless steel, copper, copper alloys, nickel-chromium alloys, super alloys and combinations thereof. In an example embodiment, the heating element 85 may be formed of nickel-chromium alloys or iron-chromium alloys. In another example embodiment, the heating element 85 may be a ceramic heater having an electrically resistive layer on an outside surface thereof.

The inner tube 70 may include a pair of opposing slots, such that the wick 90 and the first and second electrical leads 225, 225' or ends of the heating element 85 may extend out from the respective opposing slots. The provision of the opposing slots in the inner tube 70 may facilitate placement of the heating element 85 and wick 90 into position within the inner tube 70 without impacting edges of the slots and the coiled section of the heating element 85. Accordingly, edges of the slots may not be allowed to impact and alter the coil spacing of the heating element 85, which would otherwise create potential sources of hotspots. In at least one example embodiment, the inner tube 70 may have a diameter of about 4 mm and each of the opposing slots may have major and minor dimensions of about 2 mm by about 4 mm.

In at least one example embodiment, the first lead 225 is physically and electrically connected to the male threaded connector piece 155. As shown, the male threaded first connector piece 155 is a hollow cylinder with male threads on a portion of the outer lateral surface. The connector piece is conductive, and may be formed or coated with a conductive material. The second lead 225' is physically and electrically connected to a first conductive post 130. The first conductive post 130 may be formed of a conductive material

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(e.g., stainless steel, copper, etc.), and may have a T-shaped cross-section as shown in FIG. 2. The first conductive post 130 nests within the hollow portion of the first connector piece 155, and is electrically insulated from the first connector piece 155 by an insulating shell 135. The first conductive post 130 may be hollow as shown, and the hollow portion may be in fluid communication with the air passage 120. Accordingly, the first connector piece 155 and the first conductive post 130 form respective external electrical connection to the heating element 85.

In at least one example embodiment, the heating element 85 may heat pre-vapor formulation in the wick 90 by thermal conduction. Alternatively, heat from the heating element 85 may be conducted to the pre-vapor formulation by means of a heat conductive element or the heating element 85 may transfer heat to the incoming ambient air that is drawn through the e-vaping device 10 during vaping, which in turn heats the pre-vapor formulation by convection.

It should be appreciated that, instead of using a wick 90, the heating element 85 may include a porous material which incorporates a resistance heater formed of a material having a high electrical resistance capable of generating heat quickly.

As shown in FIG. 2, the second section 20 includes a power supply 145, a control circuit 185, and a sensor 190. As shown, the control circuit 185 and the sensor 190 are disposed in the housing 30'. A female threaded second connector piece 160 forms a second end. As shown, the second connector piece 160 has a hollow cylinder shape with threading on an inner lateral surface. The inner diameter of the second connector piece 160 matches that of the outer diameter of the first connector piece 155 such that the two connector pieces 155, 160 may be threaded together to form the connection 25. Furthermore, the second connector piece 160, or at least the other lateral surface is conductive, for example, formed of or including a conductive material. As such, an electrical and physical connection occurs between the first and second connector pieces 155, 160 when connected.

As shown, a first lead 165 electrically connects the second connector piece 160 to the control circuit 185. A second lead 170 electrically connects the control circuit 185 to a first terminal 180 of the power supply 145. A third lead 175 electrically connects a second terminal 140 of the power supply 145 to the power terminal of the control circuit 185 to provide power to the control circuit 185. The second terminal 140 of the power supply 145 is also physically and electrically connected to a second conductive post 150. The second conductive post 150 may be formed of a conductive material (e.g., stainless steel, copper, etc.), and may have a T-shaped cross-section as shown in FIG. 2. The second conductive post 150 nests within the hollow portion of the second connector piece 160, and is electrically insulated from the second connector piece 160 by a second insulating shell 215. The second conductive post 150 may also be hollow as shown. When the first and second connector pieces 155, 160 are mated, the second conductive post 150 physically and electrically connects to the first conductive post 130. Also, the hollow portion of the second conductive post 150 may be in fluid communication with the hollow portion of the first conductive post 130.

While the first section 15 has been shown and described as having the male connector piece and the second section 20 has been shown and described as having the female connector piece, an alternative embodiment includes the

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opposite where the first section 15 has the female connector piece and the second section 20 has the male connector piece.

In at least one example embodiment, the power supply 145 includes a battery arranged in the e-vaping device 10. The power supply 145 may be a Lithium-ion battery or one of its variants, for example a Lithium-ion polymer battery. Alternatively, the power supply 145 may be a nickel-metal hydride battery, a nickel cadmium battery, a lithium-manganese battery, a lithium-cobalt battery or a fuel cell. The e-vaping device 10 may be vapable by an adult vaper until the energy in the power supply 145 is depleted or in the case of lithium polymer battery, a minimum voltage cut-off level is achieved.

In at least one example embodiment, the power supply 145 is rechargeable. The second section 20 may include circuitry configured to allow the battery to be chargeable by an external charging device. To recharge the e-vaping device 10, an USB charger or other suitable charger assembly may be used as described below.

In at least one example embodiment, the sensor 190 is configured to generate an output indicative of a magnitude and direction of airflow in the e-vaping device 10. The control circuit 185 receives the output of the sensor 190, and determines if (1) the direction of the airflow indicates a draw on the mouth-end insert 8 (versus blowing) and (2) the magnitude of the draw exceeds a threshold level. If these vaping conditions are met, the control circuit 185 electrically connects the power supply 145 to the heating element 85; thus, activating the heating element 85. Namely, the control circuit 185 electrically connects the first and second leads 165, 170 (e.g., by activating a heater power control transistor forming part of the control circuit 185) such that the heating element 85 becomes electrically connected to the power supply 145. In an alternative embodiment, the sensor 190 may indicate a pressure drop, and the control circuit 185 activates the heating element 85 in response thereto.

In at least one example embodiment, the control circuit 185 may also include a light 60, which the control circuit 185 activates to glow when the heating element 85 is activated and/or the battery 145 is recharged. The light 60 may include one or more light-emitting diodes (LEDs). The LEDs may include one or more colors (e.g., white, yellow, red, green, blue, etc.). Moreover, the light 60 may be arranged to be visible to an adult vaper during vaping, and may be positioned between the first end 45 and the second end 50 of the e-vaping device 10. In addition, the light 60 may be utilized for e-vaping system diagnostics or to indicate that recharging is in progress. The light 60 may also be configured such that the adult vaper may activate and/or deactivate the heater activation light 60 for privacy.

In at least one example embodiment, the control circuit 185 may include a time-period limiter. In another example embodiment, the control circuit 185 may include a manually operable switch for an adult vaper to initiate heating. The time-period of the electric current supply to the heating element 85 may be set or pre-set depending on the amount of pre-vapor formulation desired to be vaporized.

Next, operation of the e-vaping device to create a vapor will be described. For example, air is drawn primarily into the first section 15 through the at least one air inlet 55 in response to a draw on the mouth-end insert 35. The air passes through the air inlet 55, into the space 250, through the air channel 270, through the central passage 235, into the inner passage 120, and through the outlet 100 of the mouth-end insert 35. If the control circuit 185 detects the vaping conditions discussed above, the control circuit 185 initiates

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power supply to the heating element **85**, such that the heating element **85** heats pre-vapor formulation in the wick **90**. The vapor and air flowing through the inner passage **120** combine and exit the e-vaping device **10** via the outlet **100** of the mouth-end insert **35**.

When activated, the heating element **85** may heat a portion of the wick **90** for less than about 10 seconds.

In at least one example embodiment, the first section **15** may be replaceable. In other words, once the pre-vapor formulation of the cartridge is depleted, only the first section **15** may be replaced. An alternate arrangement may include an example embodiment where the entire e-vaping device **10** may be disposed once the reservoir **95** is depleted. In at least one example embodiment, the e-vaping device **10** may be a one-piece e-vaping device.

In at least one example embodiment, the e-vaping device **10** may be about 80 mm to about 110 mm long and about 7 mm to about 8 mm in diameter. For example, in one example embodiment, the e-vaping device **10** may be about 84 mm long and may have a diameter of about 7.8 mm.

FIG. 3A is a cross-sectional view of a cartridge according to at least one example embodiment.

In at least one example embodiment, as shown in FIG. 3A, the first section **15** includes the transfer pad **200** that abuts the wick **90**, and the heater **85** is folded around three sides of the wick **90**.

In at least one example embodiment, the wick **90** may include one or more sheets of material, such as a sheet formed of borosilicate fibers. The sheet of material may be folded, braided, twisted, adhered together, etc. to form the wick **90**. The sheet of material may include one or more layers of material. The sheet of material may be folded and/or twisted. If multiple layers of material are included, each layer may have a same density or a different density than other layers. The layers may have a same thickness or a different thickness. The wick **90** may have a thickness ranging from about 0.2 mm to about 2.0 mm (e.g., about 0.5 mm to about 1.5 mm or about 0.75 mm to about 1.25 mm). In at least one example embodiment, the wick **90** includes braided amorphous silica fibers.

A thicker wick **90** may deliver a larger quantity of pre-vapor formulation to the heating element **85** so as to produce a larger amount of vapor, while a thinner wick **90** may deliver a smaller quantity of pre-vapor formulation to the heating element **85** so as to produce a smaller amount of vapor.

In at least one example embodiment, the wick **90** may include a stiff, structural layer and at least one additional less rigid layer. The addition of a stiff, structural layer may aid in automated manufacture of the cartridge. The stiff, structural layer could be formed of a ceramic or other substantially heat resistant material.

FIG. 3B is a perspective view of a heating element and a wick of the cartridge of FIG. 3A according to at least one example embodiment.

In at least one example embodiment, as shown in FIG. 3B, the heating element **85** may include a folded metal sheet, which at least partially surrounds the wick **90**. In at least one example embodiment, the folded heating element **85** is a single integral member that is cut and/or laser etched from a sheet of metal, which is folded about at least a portion of a wick **90**. The folded heating element **85** contacts the wick **90** on three sides.

In at least one example embodiment, the folded heating element **85** includes a first plurality of U-shaped segments arranged in a first direction and defining a first side of the heating element **85**. The folded heating element **85** also

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includes a second plurality of U-shaped segments arranged in the first direction and defining a second side of the heating element **85**. The second side is substantially parallel to the first side.

In at least one example embodiment, the folded heating element **85** also includes ends, which form a first lead portion and a second lead portion.

FIG. 4 is a graph comparing electronic vaping devices including transfer pads having different densities and/or lengths according to at least one example embodiment.

Aerosol mass of three different electronic vaping devices were compared. The first electronic vaping device (the first device) was a MARK TEN® electronic vaping device with about 0.9 g of pre-vapor formulation. The second electronic vaping device (the second device) included a cartridge having the configuration as set forth FIGS. 3A and 3B, and included an inner tube having a 1.6 mm inner diameter and the transfer pad **200** as described herein. The transfer pad had a length of about 5 mm and a density of about 0.152 g/cc. The third electronic vaping device (the third device) included a cartridge having the configuration as set forth FIGS. 3A and 3B, and included the transfer pad **200** as described herein. The transfer pad of the third electronic vaping device had a length of about 3 mm and a density of about 0.152 g/cc. The heater of each of the three devices had a resistance of about 3.5 ohms. The resistance-to-draw (RTD) of the first device was about 103 mm H₂O, the RTD of the second device was about 128 mm H₂O, and the RTD of the third device was about 129 mm H₂O.

To determine the aerosol mass, the machine smoking parameters are verified. The puff profile, puff volume, puff duration, puff time, puff frequency and number of puffs are all checked for accuracy before testing. Typical settings are: Square wave puff profile [08], puff volume 55.0 ml [64]%, puff duration 5 seconds [50], puff time 4.9 seconds [49], puff frequency 25 seconds [10], and number of puffs preset [10].

As shown in FIG. 3, the third vaping device provides a larger aerosol mass over time than the second device including the transfer pad having a longer length. Thus, delivery of the pre-vapor formulation is improved in a vaping device in which the transfer pad is shorter. Moreover, the aerosol mass of the third device was greater than the aerosol mass of the first device for about 150 puffs.

FIG. 5 is a graph comparing electronic vaping devices including transfer pads having different densities and/or lengths according to at least one example embodiment.

Aerosol mass of four different electronic vaping devices were compared. The electronic vaping device A (device A) was a MARK TEN® electronic vaping device with about 0.9 g of pre-vapor formulation, which was the control device. The electronic vaping device B (device B) included a cartridge having the configuration as set forth FIGS. 3A and 3B, but included an inner tube having a 1.6 mm inner diameter and the transfer pad **200** as described herein. The transfer pad of device B had a length of about 4 mm and a density of about 0.100 g/cc. The electronic vaping device C (device C) included a cartridge having the configuration as set forth FIGS. 3A and 3B, but included the transfer pad **200** as described herein. The transfer pad of device C had a length of about 3 mm and a density of about 0.144 g/cc. The electronic vaping device D (device D) included a cartridge having the configuration as set forth FIGS. 3A and 3B, but included the transfer pad **200** as described herein. The transfer pad of device D had a length of about 3 mm and a density of about 0.144 g/cc. The heater of each of the three devices had a resistance of about 3.5 ohms.

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The aerosol mass of each device is tested as set forth above.

As shown in FIG. 5, device B performs similarly to device D having a shorter length and higher density. Moreover, both device B and device D provide a greater aerosol mass over 140 puffs than the control device.

Accordingly, while not wishing to be bound by theory, it is believed that including a transfer pad having a shorter length and a higher density may perform similarly to an electronic vaping device including a transfer pad having a longer length and a lower density.

FIG. 6 is a photograph of cross-section of a transfer pad according to at least one example embodiment.

In at least one example embodiment, as shown in FIG. 6, the transfer pad 200 includes the plurality of fibers. When viewing the transfer pad 200, the plurality of fibers are substantially parallel.

FIG. 7 is an enlarged photograph of a transfer pad according to at least one example embodiment.

In at least one example embodiment, as shown in FIG. 7, the transfer pad 200 includes the plurality of fibers. When viewing an enlarged view of the transfer pad 200, it is shown that the plurality of fibers are substantially parallel so as to form channels between adjacent ones of the plurality of fibers.

FIG. 8 is a side view of an electronic vaping device according to at least one example embodiment.

In at least one example embodiment, as shown in FIG. 8, the electronic vaping device is the same as in FIG. 2 except that the heating element 85 is a heating coil that surrounds a portion of the wick 90.

FIG. 9 is a perspective view of a transfer pad in a cartridge, a housing of the cartridge being transparent, according to at least one example embodiment.

In at least one example embodiment, as shown in FIG. 9, the cartridge 15 is the same as in FIG. 8, except that the housing 30 of the cartridge 15 and the transfer pad 200 each have a generally square-shaped cross-section with generally rounded corners ("squirele"). The transfer pad 200 is about 3.0 mm in length so as to substantially prevent and/or reduce a weight of the transfer pad 200 from becoming too heavy when the transfer pad 200 becomes saturated with the pre-vapor formulation, and to substantially avoid and/or reduce movement of the transfer pad 200 within the cartridge 15 resulting from the weight thereof.

FIG. 10 is a perspective view of the transfer pad of FIG. 9 according to at least one example embodiment.

In at least one example embodiment, as shown in FIG. 10, the transfer pad 200 is the same as in FIG. 89, but it shown as having a length L_t of about 3.0 mm. The transfer pad 200 defines the air passage 235 therethrough. As shown, the air passage 235 has a generally circular cross-section and the cross-section of an outer surface of transfer pad 200 is different than a cross-section of the air passage 235.

FIG. 11 is a longitudinal cross-sectional view along line of an electronic vaping device according to at least one example embodiment.

In at least one example embodiment, as shown in FIG. 11, instead of a single transfer pad 200, the electronic vaping device may include cartridge and battery section having a structure that differs from the cartridge of FIGS. 1-10. The cartridge of this structure may include two or more transfer pads 200-1, 200-2.

In at least one example embodiment, as shown in FIG. 11, the electronic vaping device (e-vaping device) 1100 may include a replaceable cartridge (or first section) 1110, sometimes referred to herein as an "e-vaping tank," and a reusable

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battery section (or second section, also referred to herein as a power supply assembly) 1170, which may be coupled together. The cartridge 1110 includes a reservoir 1120 holding a pre-vapor formulation and a vaporizer assembly 1140 configured to heat pre-vapor formulation drawn from the reservoir 1120 to generate the vapor. The power supply assembly 1170 includes a power supply 1172 and is configured to, when coupled to the cartridge 1110, supply electrical power to the vaporizer assembly 1140 to enable the vaporizer assembly 1140 to generate the vapor.

The power supply assembly 1170 and the cartridge 1110 may be coupled together via respective coupling interfaces to comprise the e-vaping device 1100. The coupling interfaces may be configured to be removably coupled together, such that the power supply assembly 1170 and the cartridge 1110 are configured to be removably coupled together. It should be appreciated that each coupling interface (also referred to herein as a connector) of the coupling interfaces may include any type of interface, including a snug-fit, detent, clamp, bayonet, clasp sliding fit, sleeve fit, alignment fit, threaded connector, magnetic, clasp, or any other type of connection and/or combinations thereof. In the example embodiments shown in FIG. 11, respective inlets 1182, 1132 extend through the respective coupling interfaces to enable air to be drawn into the cartridge 1110 from the external environment ("ambient environment"). In some example embodiments, the air is drawn via an interior 1175 and an inlet 1178 of the power supply assembly 1170. The air inlet 1178 may be the only air inlet. In other example embodiments, the device may include multiple air inlets.

As shown in FIG. 11, the cartridge 1110 may include a reservoir housing 1112 at least partially defining a reservoir 1120, a vapor generator assembly 1130, and an outlet assembly 1114. As illustrated herein, the vapor generator assembly 1130 is shown to protrude from at least a portion of the reservoir 1120, but example embodiments are not limited thereto: in some example embodiments, an end of the vapor generator assembly 1130 that is distal to outlet 1118 is flush or substantially flush (e.g., flush within manufacturing tolerances and/or material tolerances) with an end of the reservoir housing 1112 that is distal to the outlet, an end of the vapor generator assembly 1130 that is proximal to outlet is flush or substantially flush with an end of the reservoir housing that is proximal to outlet, and/or vapor generator assembly 1130 may be in whole or in part within a space occupied by reservoir housing 1112. In some example embodiments, vapor generator assembly 1130 may form in whole or in part an inner tubular element of reservoir housing 1112, defining in whole or in part reservoir 1120 between outside walls of 1130 and inside walls of 1112.

In some example embodiments, a separate housing that is separate from the reservoir housing 1112 may define the vapor generator assembly 1130 and may be directly or indirectly coupled to the reservoir housing 1112. In some example embodiments, the one or more transfer pads 200-1, 200-2 may extend through both a portion of the reservoir housing 1112 and a portion of the separate housing of the vapor generator assembly 1130. In some example embodiments, the vapor generator assembly 1130 may be fixedly coupled to the reservoir housing 1112. In other example embodiments, the vapor generator assembly 1130 may be removable from and/or detachably coupled to the reservoir housing 1112. In some example embodiments, one or more seals and/or gaskets may be between the coupled housings.

The cartridge 1110 may include a structural element (also referred to herein as an inner tube 1122) within a space at least partially defined by the reservoir housing 1112. The

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reservoir housing 1112 and the inner tube 1122 may each be configured to at least partially define the reservoir 1120. For example, an inner surface of reservoir housing 1112 may define an outer boundary of reservoir 1120. In another example, as shown, an outer surface of inner tube 1122 may define an inner boundary of reservoir 1120. As shown, the reservoir 1120 may be defined as a space between an outer surface of inner tube 1122 and an inner surface of reservoir housing 1112. In some example embodiments, vapor generator assembly 1130, in whole or in part, may form a part of inner tube 1122.

In some example embodiments, cap structure 1198 may be coupled to ends of reservoir housing 1112 and inner tube 1122 that are proximal to outlet 1118 and thus complete the enclosure of the reservoir 1120. As shown, cap structure may be further coupled to an outlet structure 1114, and cap structure 1198 may include a port extending therethrough which is configured to enable fluid communication between the interior of inner tube 1122 (e.g., channel 1124) and channel 1116 of outlet structure 1114. In some example embodiments, cap structure 1198 may be fixedly coupled to outlet structure 1114 and/or to reservoir housing 1112. In some example embodiments, cap structure 1198 may be detachably coupled to the cap structure 1198, thereby enabling the outlet structure 1114 to be coupled or detached from a remainder of the e-vaping device 100 without further exposing the reservoir 120. In some example embodiments, reservoir housing 112 and inner tube 122 can be parts of a unitary piece (i.e., parts of a single piece). In some example embodiments, reservoir housing 112 and cap structure 1198 can be parts of a unitary piece. In some example embodiments, inner tube 122 and cap structure 1198 can be parts of a unitary piece. In some example embodiments, cap structure 1198, reservoir housing 112 and/or inner tube 122 can be individual parts coupled together, or parts of a unitary piece.

In further example embodiments, cap structure 1198 and outlet structure 1114 can be parts of a unitary piece. In further example embodiments, cap structure 1198 and reservoir housing 112 can be parts of a unitary piece. In other words, cap structure 1198 may simply be a part of outlet structure 1114 or of reservoir housing 112, or all may be parts of the same unitary piece. In yet further example embodiments, outlet structure 1114, cap structure 1198, reservoir housing 112 and/or inner tube 112 can be individual parts coupled together, or parts of a unitary piece.

The inner surface of inner tube 1122 at least partially defines a channel 1124. As shown in FIG. 11, the inner tube 1122 may extend through at least one end of the reservoir housing 1112 so that the channel 1124 is in fluid communication with vaporizer assembly 1140 within an interior of the vapor generator assembly 1130, an inlet 1132, and outlet 1118.

Still referring to FIG. 11, the vapor generator assembly 1130 includes a vaporizer assembly 1140 configured to draw pre-vapor formulation from the reservoir 1120 and to heat the drawn pre-vapor formulation to generate a vapor. The vaporizer assembly 1140 may include one or more transfer pads 200-1, 200-2 that extend through at least one structure that at least partially defines the reservoir 1120. In some example embodiments, transfer pads 200-1, 200-2 may also extend through at least one structure that at least partially defines the vapor generator assembly 1130. In some embodiment, transfer pads 200-1, 200-3 may also extend into the vapor generator assembly 1130, such that a first part of transfer pad 200-1 and a first part of transfer pad 200-2 reside inside reservoir 1120, and also a second part of

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transfer pad 200-1 and a second part of transfer pad 200-2 reside inside the vapor generator assembly 1130. In some example embodiments, as shown in FIG. 11, vapor generator assembly 1130 includes transfer pads 200-1, 200-2 that extend through an end portion 1115 of reservoir housing 1112, where the end portion 1115 of reservoir housing 1112 at least partially defines the end of reservoir 1120 and an end of the generator assembly 1130, so that the transfer pads 200-1, 200-2 are in fluid communication with the reservoir 1120. Each transfer pad 200-1, 200-2 is configured to draw pre-vapor formulation from the reservoir 1120 at the respective ends that are inside the reservoir, and through an interior of the respective transfer pads 200-1, 200-2 to respective opposite ends thereof. In some example embodiments, including the example embodiments shown in at least FIG. 11, the transfer pads 200-1, 200-2 may be cylindrical in shape, but it will be understood that other shapes and sizes of the transfer pads 200-1, 200-2 may be possible (for example, the transfer pads may be flat and/or may have other cross-sectional forms such as square, rectangular, oval, triangular, irregular, others, etc. and/or combinations thereof). Each transfer pad may also have a different shape. In some example embodiments, one or more of the transfer pads may have a cylindrical shape such that the one or more transfer pads is about 2.5 mm in diameter and about 4.0 in height. Any other dimensions may be used depending on the application. In some example embodiments, one or more of the transfer pads may at least partially comprise polyethylene terephthalate (PET), polypropylene (PP), a mixture of PET and PP, or the like. In some example embodiments, the transfer pads may be made of any materials with capabilities to transfer pre-vapor formulation from one location to another either through wicking or through other mechanisms. In some example embodiments, only one transfer pad may be used, or more than two transfer pads may be used.

The vaporizer assembly 1140 further includes a dispensing interface 1144 (e.g., a "wick") and a heating element 1142. The dispensing interface 1144 is in contact with the respective ends of the one or more transfer pads 200-1, 200-2, such that pre-vapor formulation drawn from the reservoir 1120 by the one or more transfer pads 200-1, 200-2 may be drawn through the one or more transfer pads 200-1, 200-2 to the dispensing interface 1144. Thus, the dispensing interface 1144 may draw pre-vapor formulation from the reservoir 1120 via the one or more transfer pads 200-1, 200-2 (as noted above, less or more transfer pads may be used). The heating element 1142 is configured to generate heat that heats the pre-vapor formulation drawn into the dispensing interface 1144 from the reservoir 1120 via the one or more transfer pads 200-1, 200-2. In some example embodiments, the heating element 1142 is in contact with the dispensing interface 1144. In some example embodiments, the heating element is isolated from direct contact with the dispensing interface 1144. In some example embodiments, one or more transfer pads and the dispensing interface can be individual parts that contact each other, or can be parts of a unitary piece. In some example embodiments, the heating element 1142 may be on (e.g., may at least partially cover) each side of opposite sides of the dispensing interface 1144. In some example embodiments, the heating element 1142 may at least partially extend around (e.g., may at least partially wrap around) the dispensing interface.

In some example embodiments, the vapor generator assembly 1130 includes a circuit 1148 and an interface 1149 that is configured to couple with an interface 1180 of the power supply assembly 1170. The interface 1149 is config-

ured to electrically couple the vaporizer assembly 1140 and the circuit 1148 with the power supply assembly 1170 via interface 1180 of the power supply assembly 1170.

In some example embodiments, including the example embodiments shown in FIG. 11, the interior of vapor generator assembly 1130 is at least partially defined by the same reservoir housing 1112 that at least partially defines the reservoir 1120. In some example embodiments, the interior of vapor generator assembly 1130 is at least partially defined by a different housing relative to reservoir housing 1112, such that reservoir housing 1112 does not at least partially define the interior of vapor generator assembly 1130.

The interface 1149 includes an inlet 1132 that extends through the interface 1149, and may at least partially extend through the circuit 1148, so that the vaporizer assembly 1140 in the interior of vapor generator assembly 1130 is in fluid communication with an exterior of the cartridge 1110 via the inlet 1132. As shown in FIG. 11, an end of the inner tube 1122 may be in fluid communication with an interior of the vapor generator assembly 1130 and thus may be in fluid communication with the vaporizer assembly 1140 located within the vapor generator assembly 1130. Air entering the cartridge 1110 via inlet 1132 may flow through the interior of the vapor generator assembly 1130, in fluid communication with the vaporizer assembly 1140, to flow into channel 1124 defined by inner tube 1122.

Referring to FIG. 11, the e-vaping device 1100 includes electrical pathways that may electrically couple the heating element 1142 to the interface, thereby enabling the heating element 1142 to be electrically coupled to power supply 1172 based on the interface of cartridge 1110 being coupled with the interface of power supply assembly 1170. The electrical pathways may include one or more electrical connectors.

If and/or when the interfaces are coupled together, one or more electrical circuits ("pathways") through the cartridge 1110 and the power supply assembly 1170 may be established ("closed"). The established electrical circuits may include the vaporizer assembly 1140, electrical pathways, circuit 1148, interfaces, control circuitry 1176, power supply 1172, sensor 1174, light source 1177 (e.g., a light-emitting diode ("LED")), and/or the one or more light-emitting devices 1188-1 and 1188-2. As described further herein, the light source 1177 and/or the one or more light-emitting devices 1188-1 and 1188-2 are configured to emit light having a selected one or more properties ("light properties") of a plurality of properties (e.g., a selected color of a plurality of colors, a selected brightness of a plurality of brightness levels, a selected pattern of a plurality of patterns, a selected duration of a plurality of durations, some combination thereof, or the like).

Referring now to the outlet assembly 1114 as shown in FIG. 11, the outlet assembly 1114 includes a channel 1116 extending therethrough to establish the outlet 1118. In example embodiments where cap structure 1198 is simply a part of outlet assembly 1114, or a part of reservoir housing 1112, or where cap structure 1198 is omitted from the cartridge 1110, the outlet assembly 1114 may be coupled, at an end thereof, to reservoir housing 1112 and/or inner tube 1122 to couple channel 1116 with inner tube 1122, thereby enabling vapor to flow through the channel 1124 of the inner tube 1122 to the channel 1116 to the outlet 1118. In example embodiments where cap structure 1198 is a separate piece in between outlet assembly 1114 and reservoir housing 1112, the outlet assembly 1114 may be coupled to cap structure 1198 at an end of cap structure 1198, and cap structure 1198

will be coupled on an opposing end to reservoir housing 1112 and/or inner tube 1122, to enable vapor to flow through to the outlet 1118.

Referring now to cartridge 1110 as a whole, in view of the above, the cartridge 1110 may be configured to receive a flow of air into the vapor generator assembly 1130 via inlet 1132, generate a vapor at vaporizer assembly 1140, enable the generated vapor to be entrained in the flow of air through the interior of vapor generator assembly 1130, direct the flow of air with generated vapor into the channel 1124 from the vapor generator assembly 1130, and direct the flow of air with generated vapor to flow through the channel 1124 and channel 1116 (and through 1198a if 1198 is a separate piece in between 1112 and 1114) to the exterior environment via outlet 1118.

FIG. 12 is an exploded view of the cartridge of FIG. 11 according to at least one example embodiment.

In at least one example embodiment as shown in FIG. 12, the cartridge 1110 is generally the same as in FIG. 11, but the transfer pads 200-1, 200-2 are shown as being generally cylindrical bodies. Further, as shown in FIG. 12, cap structure 1198 may be used to at least partially seal an open end of the reservoir housing 1112.

In other example embodiments, not shown, the housing of the electronic vaping device and/or the at least one transfer pad 200 may have other cross-sectional shapes, including triangular, rectangular, oval, or any other suitable shape.

To test the cartridges for leaks, the following procedure was followed to determine t_0 (time before incubation), t_{end} (time after incubation), $\Delta_{weight} (mg) = wt_{0(mg)} - wt_{end(mg)}$, and $\Delta_{weight} (\%) = ((wt_{0(mg)} - t_{end(mg)}) / \text{fill weight}) * 100$. The test utilized a vacuum oven (VO-200 Memmert or equivalent with a vacuum pump) and an analytical balance (OHAUS PA214C or equivalent).

To determine leakage, empty cartridges are weighed together with the parts used for assembly if needed (e.g. mouthpiece, sealing ring). Each cartridge is filled with a pre-vapor formulation. The full Cartridges are assembled if needed (e.g. mouthpiece, sealing ring) and weighed. The full Cartridges stand for about 30 minutes at least prior to analysis. The full cartridges are then sealed inside foil bag and inserted into the Vacuum Oven: Ambient/500 mbar. The cartridges are incubated for about 24 hours. Then the foil bag is opened and visually inspected for drops. Each cartridge is then wiped down and weighed. Each cartridge is then puffed 50 puffs (5 second draw/55 ml, 30 sec total). The cartridges are then incubated at about 55° C. overnight in a horizontal position. The cartridges are then wiped and then weighed.

When tested for leakage as set forth above, the cartridge of FIGS. 3A and 3B, including a transfer pad having a density of 0.100 g/cc and no gasket between the liquid reservoir and the heating element and wick showed no leaks. As compared to control cartridge including no transfer pad, but including a seal/gasket between the reservoir and the heating element, the cartridge of FIGS. 3A and 3B performed substantially the same as the control cartridges as shown in Table 1 below.

TABLE 1

Tank No.	Orientation	Before Oven Weight (g)	After Oven Weight (g)	Weight Difference (g)	Ratio of Weight Change (%)
1	Horizontal	8.1735	8.1741	0.0006	0.01%
2	Horizontal	8.1658	8.1671	0.0013	0.02%

TABLE 1-continued

Tank No.	Orientation	Before Oven Weight (g)	After Oven Weight (g)	Weight Difference (g)	Ratio of Weight Change (%)
3	Horizontal	8.1631	8.1639	0.0008	0.01%
C1	Horizontal	8.4910	8.4932	0.0022	0.03%
C2	Horizontal	8.4631	8.4645	0.0014	0.02%
C3	Horizontal	8.4952	8.4968	0.0016	0.02%

Example embodiments have been disclosed herein, it should be understood that other variations may be possible. Such variations are not to be regarded as a departure from the spirit and scope of the present disclosure, and all such modifications as would be obvious to one skilled in the art are intended to be included within the scope of the following claims.

We claim:

1. A cartridge for an electronic vaping device comprising:
 - an outer housing extending in a longitudinal direction;
 - a reservoir configured to contain a pre-vapor formulation, the reservoir in the outer housing, the reservoir including,
 - a reservoir housing in the outer housing and defining two orifices therein;
 - at least two transfer pads at a reservoir end, the transfer pads including a plurality of fibers, each of the plurality of fibers being substantially parallel to the longitudinal direction, and each of the transfer pads extending through a respective one of the two orifices in the reservoir housing;
 - a cap structure coupled to the reservoir housing opposite the reservoir end and defining a port;
 - an outlet assembly coupled to the cap structure and defining a channel, the channel fluidly coupled to the port of the cap structure; and
 - a vapor generator assembly in contact with the transfer pads, the vapor generator assembly including,
 - a wick in contact with the transfer pads, and
 - at least one heater in fluid communication with the wick, the vapor generator assembly configured to direct air flow in the longitudinal direction across the at least one heater.
2. The cartridge of claim 1, wherein the reservoir is under atmospheric pressure.
3. The cartridge of claim 1, wherein the transfer pads have a density ranging from about 0.08 g/cc to about 0.3 g/cc.
4. The cartridge of claim 1, wherein the transfer pads have a length of about 2.0 mm to about 10.0 mm.
5. The cartridge of claim 1, wherein the transfer pads have a density of about 0.08 g/cc to about 0.1 g/cc.
6. The cartridge of claim 1, wherein the transfer pads have a length of about 0.5 mm to about 10.0 mm.
7. The cartridge of claim 1, wherein the transfer pads have a density of about 0.1 g/cc to about 0.3 g/cc.
8. The cartridge of claim 1, wherein the transfer pads have a length of about 3.0 mm.
9. The cartridge of claim 1, wherein the transfer pads have a generally cylindrical cross-section.
10. The cartridge of claim 1, wherein the transfer pads each include a plurality of channels, one or more of the plurality of channels between adjacent ones of the plurality of fibers.
11. The cartridge of claim 1, wherein each of the transfer pads includes an outer side wall, the outer side wall having a coating thereon.

12. The cartridge of claim 1, further comprising:

an inner tube within the outer housing, the reservoir between an outer surface of the inner tube and an inner surface of the outer housing.

13. The cartridge of claim 1, wherein the plurality of fibers include polypropylene, polyester, or both polypropylene and polyester.

14. The cartridge of claim 1, wherein the at least one heater does not contact the transfer pads.

15. The cartridge of claim 1, wherein about 50% to about 100% of the plurality of fibers extend substantially in the longitudinal direction.

16. The cartridge of claim 15, wherein about 75% to about 95% of the plurality of fibers extend substantially in the longitudinal direction.

17. An electronic vaping device comprising:

an outer housing extending in a longitudinal direction;

a reservoir configured to contain a pre-vapor formulation, the reservoir in the outer housing, the reservoir including,

a reservoir housing in the outer housing and defining two orifices therein;

at least two transfer pads at a reservoir end, each of the transfer pads including a plurality of fibers, the plurality of fibers being substantially parallel to the longitudinal direction, and each of the transfer pads extending through a respective one of the two orifices in the reservoir housing;

a cap structure coupled to the reservoir housing opposite the reservoir end and defining a port;

an outlet assembly coupled to the cap structure and defining a channel, the channel fluidly coupled to the port of the cap structure;

a vapor generator assembly in contact with the transfer pads, the vapor generator assembly including,

a wick in contact with the transfer pads;

at least one heater in fluid communication with the wick, the vapor generator assembly configured to direct air flow in the longitudinal direction across the at least one heater; and

a power supply electrically connectable to the at least one heater.

18. The electronic vaping device of claim 17, wherein the reservoir is under atmospheric pressure.

19. The electronic vaping device of claim 17, wherein the transfer pads have a density ranging from about 0.08 g/cc to about 0.3 g/cc.

20. The electronic vaping device of claim 17, wherein the transfer pads have a length of about 2.0 mm to about 10.0 mm.

21. The electronic vaping device of claim 17, wherein the transfer pads have a density of about 0.08 g/cc to about 0.1 g/cc.

22. The electronic vaping device of claim 17, wherein the transfer pads have a length of about 0.5 mm to about 5.0 mm.

23. The electronic vaping device of claim 17, wherein the transfer pads have a density of about 0.1 g/cc to about 0.3 g/cc.

24. The electronic vaping device of claim 17, wherein the transfer pads include a plurality of channels, one or more of the plurality of channels between adjacent ones of the plurality of fibers.

25. The electronic vaping device of claim 17, wherein the transfer pads include an outer side wall, the outer side wall having a coating thereon.

26. The electronic vaping device of claim 17, wherein the transfer pads have a length of about 3.0 mm.

27. The electronic vaping device of claim 17, wherein the transfer pads have a generally cylindrical cross-section.

28. The electronic vaping device of claim 17, further comprising:

an inner tube within the outer housing, the reservoir between an outer surface of the inner tube and an inner surface of the outer housing.

29. The electronic vaping device of claim 17, wherein the plurality of fibers include polypropylene, polyester, or both polypropylene and polyester.

30. The electronic vaping device of claim 17, wherein the at least one heater does not contact the transfer pads.

31. The electronic vaping device of claim 17, wherein about 50% to about 100% of the plurality of fibers extend substantially in the longitudinal direction.

32. The electronic vaping device of claim 17, wherein about 75% to about 95% of the plurality of fibers extend substantially in the longitudinal direction.

33. The cartridge of claim 1, wherein, the vapor generator assembly further includes an interface configured to be coupled to a power supply, and

a first end of the at least two transfer pads is configured to extend into the reservoir and a second end of the at least two transfer pads is configured to extend into the vapor generator assembly.

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